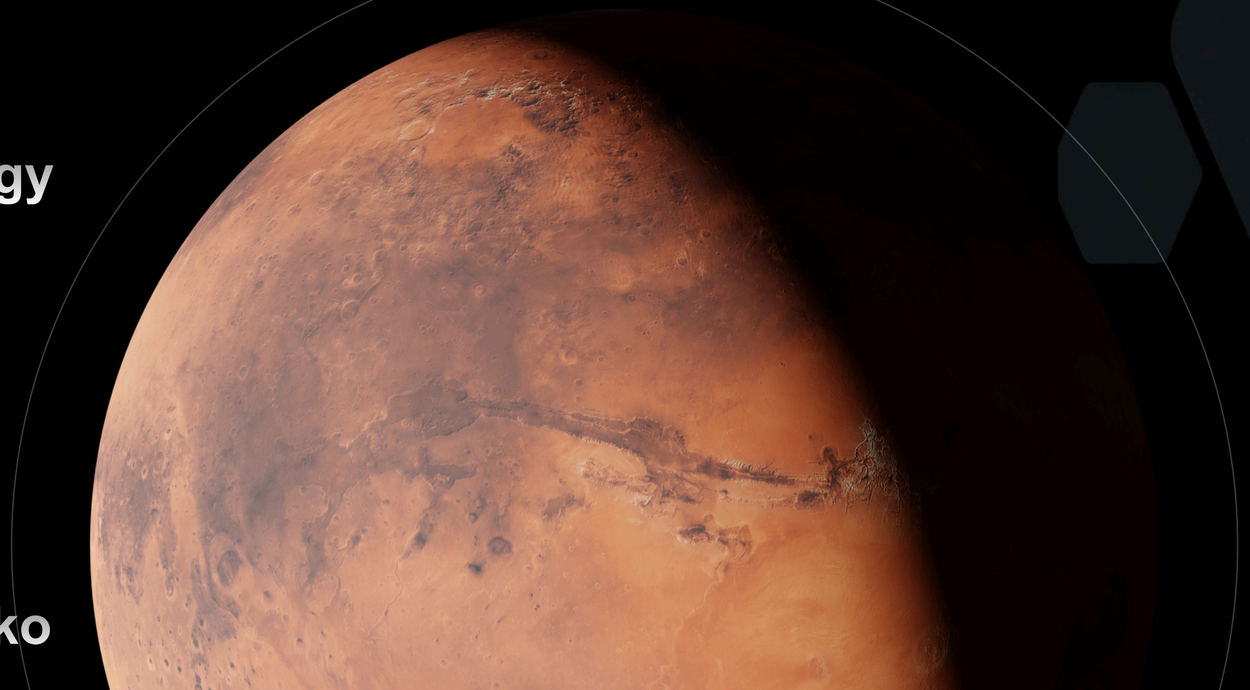
SMART INDIA HACKATHON 2026

PROJECT ANGARAKA

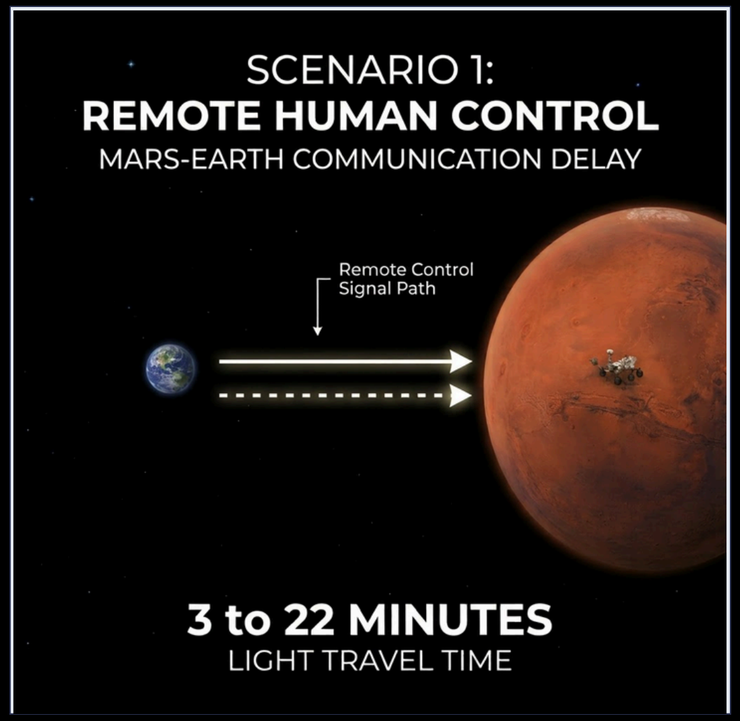
- **Problem Statement ID - SIH26226**
- **Problem Statement Title - Student Innovation-Space technology refers to the application of engineering principles to the design, development, manufacture, and operation of devices and systems for space travel and exploration.**
- **Theme - Space Technology**
- **PS Category - Hardware**
- **Team ID - 126234**
- **Team Name - Maneki_Neko**



Maneki_Neko

PROJECT ANGARAKA

SMART INDIA HACKATHON 2026



THE PROBLEM

A SOL is 1 Martian day - approximately 24 hours 39 minutes. Each day's activities are prepared on Earth and transmitted to the rover across a 3–22 minute light delay.

Rovers can already avoid obstacles, but their goals are still chosen on Earth. When a rover encounters unexpected terrain or a new opportunity, it cannot simply ask Earth and wait.

A decision that takes seconds on Mars can take minutes to communicate, forcing the rover to act on a plan made before the latest situation was known.



PROPOSED SOLUTION

An onboard small language model chooses the rover's next goal. Proven planners and a learned driving RL policy carry it out.

HOW IT ADDRESSES THE PROBLEM

Decisions are made on Mars in seconds, not on Earth a sol later. Decisions which require immediate actions face almost no delay.

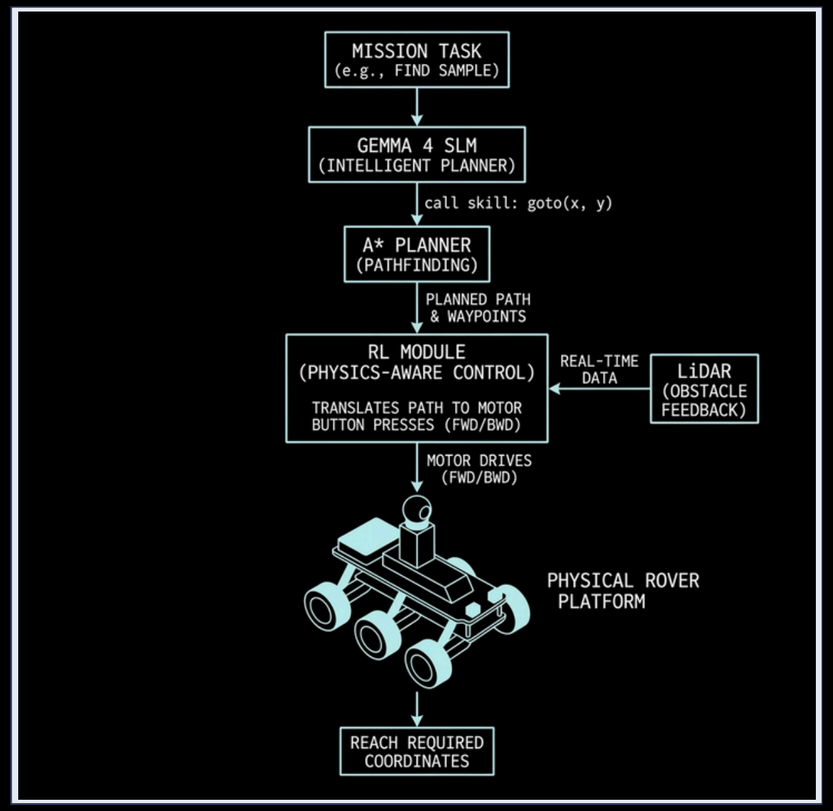
INNOVATION AND UNIQUENESS

The SLM never drives. It has knowledge and no hands; the RL policy has hands and no knowledge. Both fill others gaps.

Maneki_Neko

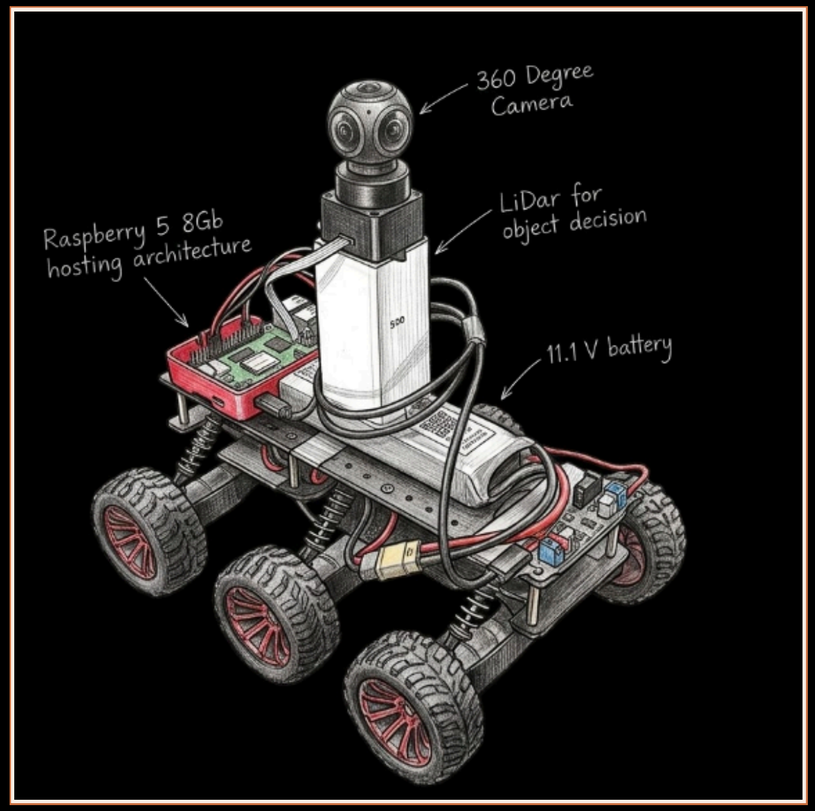
TECHNICAL APPROACH

SMART INDIA HACKATHON 2026



METHODOLOGY: THE REFLEX ARC

The SLM decides what is worth doing and calls a skill, goto(x, y). A* plans the route, and a learned RL policy drives it, corrected by LiDAR. Thinking takes seconds, planning takes milliseconds, and execution by RL happens in real time.



TECHNOLOGIES USED

A Raspberry Pi 5 (8 GB) runs the whole stack. The 360° camera finds objectives and discovers unexplored ground; LiDAR tracks position and heading, measuring exact coordinates and returning errors.

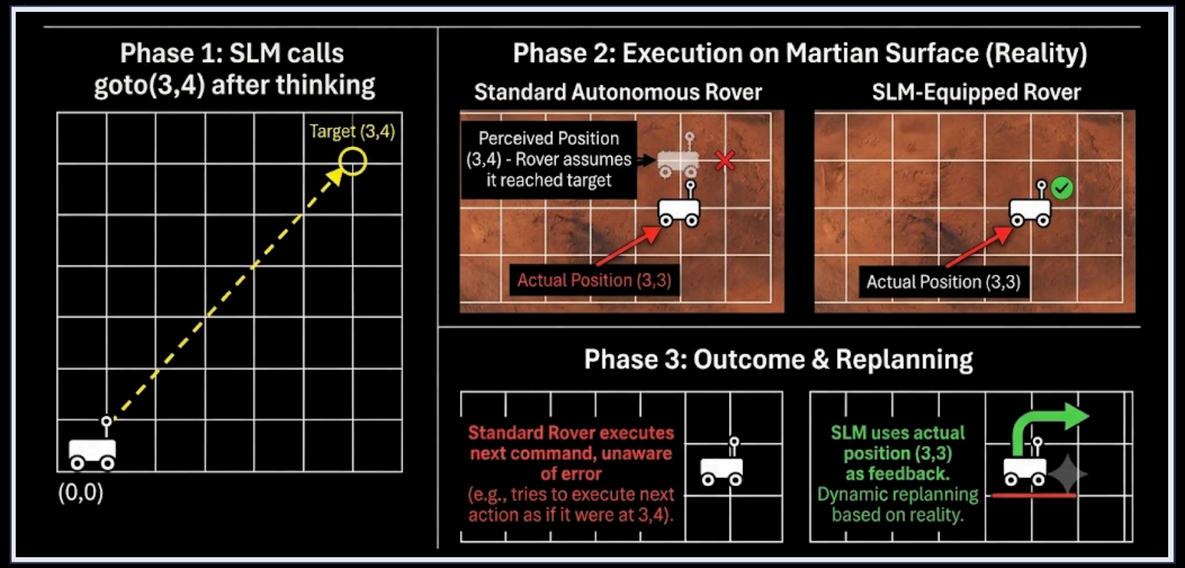
Python · Gemma 4 · A* · RL (Unity) · 6-wheel chassis · SmartElex 15D driver · 11.1 V LiPo



WORKING PROTOTYPE

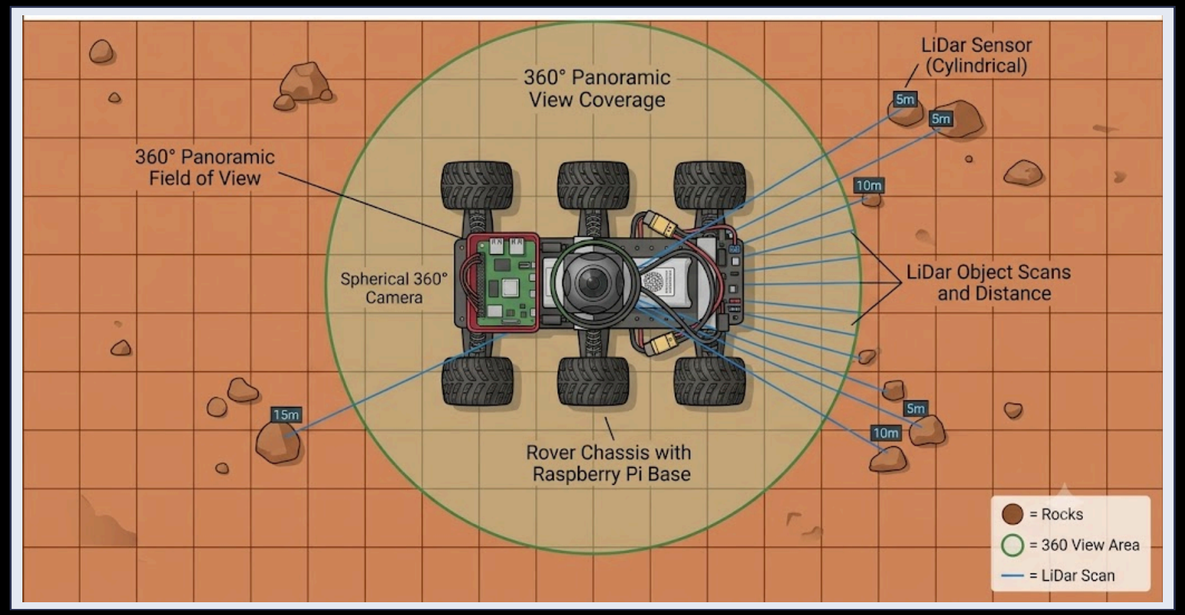
Our simulation: the SLM explores a fogged Martian map alone. It calls skill tools, completes objectives, returns to base, and notes what it found for the next sol. The white dot is the rover, brightened tiles around it is the 360° camera and the SLM thinking and conversation is on the right.

FEASIBILITY AND VIABILITY



FEASIBILITY

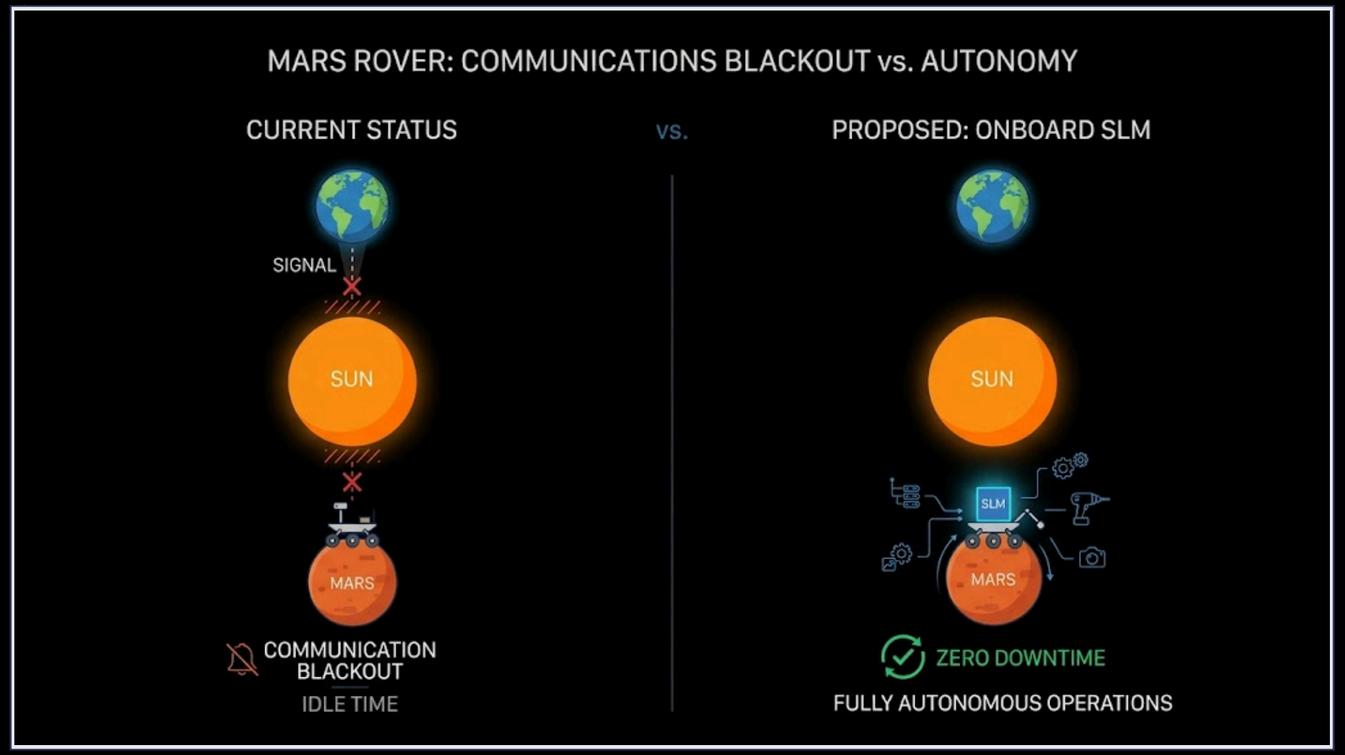
- Self-driving already works on Mars: Perseverance's AutoNav steers around hazards without help from Earth. Angaraka adds the layer above it, choosing the goal.
- It decides a few times a sol, not every second, so a slow onboard chip is fast enough.
- The model never touches the motors. Every goal goes through A* and a trained policy, so a bad decision costs a detour, not the rover.
- Tried and tested - by giving the SLM tools like a scratchpad and a memory document, it is able to carry forward context cheaply across multiple days without ever poisoning its context.



CHALLENGES → STRATEGIES

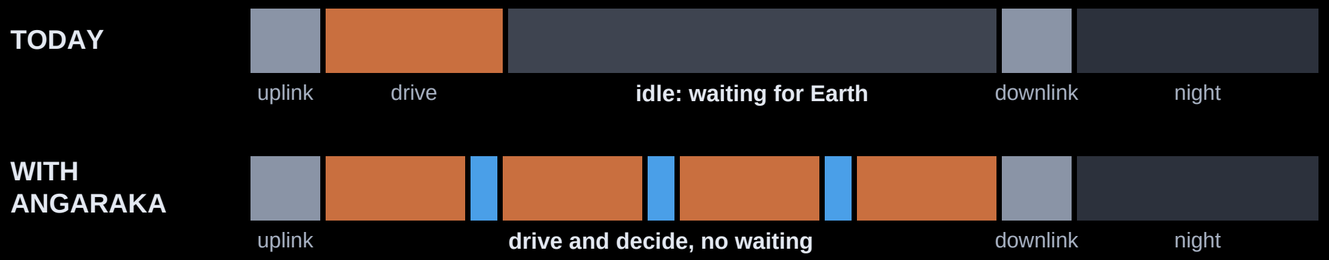
- ONBOARD COMPUTE: today's rover computer is a 200 MHz RAD750 with 256 MB of RAM, too small for a language model. → NASA and Microchip's HPSC processor has over 100× today's space computing and is built for onboard AI. Ingenuity has already flown a commercial phone chip on Mars.
- POSITION DRIFT: told to go to (3,4), the rover stops at (3,3). → LiDAR re-measures after every leg, and the SLM replans from the true position. Errors get reported, not hidden.
- UNEXPLORED GROUND: the SLM can't plan around what it hasn't seen. → The 360° camera scans all around, clearing the fog and finding objectives as the rover moves.

IMPACT AND BENEFITS



Solar conjunction: about two weeks every two years with no commands from Earth. Today the rovers park. With an onboard SLM the work continues, and the data comes home when the link returns.

ONE SOL WITH AND WITHOUT ANGARAKA



proportions illustrative, not to scale

IMPACT ON THE TARGET AUDIENCE

Mission teams at ISRO, NASA and ESA

- Teams set intent (“survey this crater”) instead of scripting every move.
- Scientists get targets chosen from what the rover saw today, not yesterday’s images.
- Idle hours become driving hours, and conjunction no longer means parking.

BENEFITS

- SCIENTIFIC: more science from every sol of a rover’s limited life.
- ECONOMIC: more return from a fixed-cost mission, and lighter daily operations. JPL’s own study argues onboard computing cuts operations cost.
- SOCIAL: open source on a Raspberry Pi, so a student team can reproduce it.
- ENVIRONMENTAL: the same stack fits places on Earth with poor communication, such as disaster sites, mines, and polar or deep-sea surveys.
- SCALES WITH DISTANCE: Mars is minutes away, the outer planets are hours. The longer the link, the more it’s worth.

RESEARCH AND REFERENCES

THE PROBLEM: NASA / JPL

- AutoNav: self-driving Perseverance ‘takes the wheel’ (JPL, 2021)
- Mars solar conjunction: the fleet lies low (JPL, 2021)
- Perseverance rover components: the RAD750 computer (NASA)
- High Performance Spaceflight Computing, HPSC (NASA)
- Cretu et al.: Snapdragon SoC on Ingenuity (NTRS, 2022)
- Didier et al.: rover ops with onboard computing (NTRS, 2022)

THE ARCHITECTURE

- SayCan: language grounded in robot affordances (arXiv:2204.01691)
- MLNav (JPL): learned proposer, safe verifier (arXiv:2203.04563)
- Hösch et al.: LLM choosing among RL skills (arXiv:2606.20014)
- LARAP: LLM over robot action primitives (Scientific Reports, 2025)
- SCALAR: skills with preconditions, Craftax (arXiv:2603.09036)

OURS

- github.com/IshuIsAwake/reflex-arc
- https://drive.google.com/drive/folders/1j6fbdQPfck_icCluUQJ67qGyCbwjTMF3?usp=sharing



▶ SIMULATION: the SLM explores and completes objectives making smart decisions independently and answering questions



▶ ROVER: LiDAR maps the surroundings and tracks the rover’s own movement ·